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16. Tree of Life

Content

- Taxonomy: Science of Classification, Milestones.
- Levels of classification by Carl Linnaeus.
- Five Kingdom classification of Robert H. Wittaker
- Binomial Nomenclature
- Modern Classification – Carl Woese' Six Kingdoms

Questions & Answers

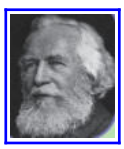
1. Taxonomy?
   Taxonomy is the branch of biology that identifies organisms, classifies them on the basis of similarities and differences and names them scientifically.
2. Basis of classification?
   Organisms are classified into different groups based on similarities and differences in their structure and genetic make-up.
3. Who is known as the 'Father of Taxonomy' and why?
   Carl Linnaeus, who fixed taxonomic hierarchy levels (*taxa*) and provided a scientific base for classification as well as scientific naming (*binomial nomenclature*).

4. Scientists who tried to classify organisms:

Aristotle	BC (Greece)	Classified organisms as those with red blood cells and those without.
Theophrastus	BC (Greece)	Classified plants into **Trees**, **Shrubs** and **Herbs**.
John Ray	17th century (England)	- Initiated the scientific method of classification. - Used the term 'species' for the first time.
Carl Linnaeus 'Father of Modern Taxonomy'	18th century (Sweden)	- Classified organisms into **2** kingdoms: Animalia & Plantae. - Introduced different **levels of classification** and provided scientific foundation for classification. - Proposed scientific naming, Binomial nomenclature.
Ernst Heaeckel	19th century	Classified living organisms into **3** kingdoms, as Animalia, Plantae & Protista
Robert H. Whittaker	America	Classified organisms in **5** kingdoms, as **Monera, Protista, Fungi, Plantae** and **Animalia.**
Carl Woese	America	Included organisms in **6** kingdoms, as Archaebacteria, Eubacteria, Protista, Fungi, Plantae and Animalia.









5. Which are the dfferent **Levels of Classification** or **Taxa** proposed by Carl Linnaeus ?
   - Kingdom
   - Phylum /Division
   - Class
   - Order
   - Family
   - Genus
   - Species

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6. The most basic level of classification?
   Species. (in 17th century, John Ray introduced this term)

7. Define a 'species'?
   A species is a group of organisms that can naturally reproduce fertile offspring.
   Eg: Human being, cat, dog, cashew tree, coconut tree …

8. The classification levels (taxa) of tiger (*Panthera tigris*):

Taxa / Levels	Taxa Group	Characteristics
Kingdom	**Animalia**	(Animals) -Multicellular, eukaryotic, heterotrophic organisms.
Phylum	**Chordata**	(Vertebrates) -Presence of vertebrae and vertebral column.
Class	**Mammalia**	(Mammals) -Warm-blooded, body covered with hair, nourish young ones with milk.
Order	**Carnivora**	(Carnivores) -Sharp teeth and claws adapted for hunting and eating meat.
Family	**Felidae**	Retractable claws, excellent night vision.
Genus	***Panthera***	(Big cats) -Ability to roar, large and powerful body structure.
Species	***tigris***	Orange fur with black stripes and strong large body.

9. The classification levels (taxa) of coconut tree (*Cocos nucifera*):

Taxa / Levels	Taxa Group	Characteristics
Kingdom	**Plantae**	(Plants) -Multicellular, eukaryotic, autotrophic organisms.
Division	**Magnoliophyta**	(Flowering plants) -Seeds enclosed inside fruits.
Class	**Liliopsida**	(Monocotyledons) Plants with parallel leaf venation and fibrous roots.
Order	**Arecales**	Plants with long and hard stems.
Family	**Arecaceae**	With large compound leaves and unbranched stems.
Genus	***Cocos***	Produces large, single-seeded fruits.
Species	***nucifera***	Fruits with a hard shell, fibrous husk, with a liquid and solid endosperm, inside.

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10. Robert H. Wittaker's Five Kingdoms classification?

Kingdom **Monera**	**Prokaryotes**, do not have a distinct nucleus. They are simplest and most ancient living organisms.	**Eg:**- Bacteria
Kingdom **Protista**	Most of them are **unicellular eukaryotes**. Autotrophs or heterotrophs. Most of them are live in aquatic habitats. Flagella, cilia, pseudopodia etc are their locomotory structures.	Amoeba, Paramecium, Euglena
Kingdom **Fungi**	Eukaryotes that are **saprophytes**. They do not have chlorophyll. Majority of them are unicellular.	Mould,Yeast, Mushroom
Kingdom **Plantae**	Multicellular eukaryotes that produce their own food through photosynthesis (**Autotrophs**). They are primary producers. Have roots, stems and leaves. They grow in soil or water.	Trees, Herbs.
Kingdom **Animalia**	Multicellular eukaryotes and **heterotrophs**. They have locomotory organs and definite receptors.	Sponge, Man

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11. Why Robert H. Whittaker classified all organisms into five kingdoms?
Carl Linnaeus classified all organisms into two kingdoms (Animalia & Plantae). But Bacteria, Paramecium, Mushroom and Yeast cannot be included in either the Plant or Animal kingdoms. To include these, Robert H. Whittaker classified organisms into five kingdoms, as Monera, Protista, Fungi, Plantae and Animalia.

12. Thoug a few bacteria in the Monera kingdom are disease causing pathogens, majority of them are highly useful. Cite any two examples?
Bacteria play an important role in nutrient cycles like nitrogen fixation in the soil.
Also helpful in our digestion process.

13. Identify and name these organisms. In which kingdom they belong?



Euglena
Amoeba
Paramecium. All are included in the kingdom Protista.

14. Name any two protists that cause diseases.
Plasmodium, *Entamoeba histolytica.*

15. Why fungi are known as saprophytes?
Since fungi do not have chlorophyll, they cannot perform photosynthesis. They absorb nutrients from dead and decaying matter, so they are called saprophytes.

16. Name a unicellular fungus.
Yeast.

17. The cell wall of fungus is made of a substance called, ------- ?
Chitin.

18. 'There are useful and harmful fungi around us'. Justify this statement.
Some mushrooms are edible food items. Yeast is used in baking and preparation of wine and alcohol. But some other fungi cause diseases such as candidiasis and ringworm.

19. What is Binomial nomenclature? Who proposed this?
The scientific method of naming where the genus name comes first and the species name second is called Binomial nomenclature.
Carl Linnaeus is the proponent of this scientific naming method.

20. The group which is included organisms of similar species, is a ---------- ?
Genus.

21. Why there is a scientific system of naming organisms?
An organism may have different names according to regions and languages.
Binomial nomenclature is useful to avoid such confusions.

22. The rules of scientific naming system?
- The name of an organism should have two parts. The first part is the genus name and second part is the species name.
- When writing in English, the genus name should begin with a capital letter.
- The scientific name should be printed in italics.

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23. Give the scientific names of a few organisms that are seen around us?

Cat:	*Felis domesticus*	Coconut tree:	*Cocos nucifera*
Dog:	*Canis familiaris*	Paddy	: *Oryza sativa*
Tiger:	*Panthera tigris*	Mango tree	: *Mangifera indica*
Lion:	*Panthera leo*	Neem	: *Azadirachta indica*
Man:	*Homo sapiens*	Hibiscus	: *Hibiscus rosasinensis*

24. What is the main reason classification systems need to be continuously updated?
Classification systems need to be updated because new organisms are discovered regularly, and the existing systems may be insufficient to accurately categorize them. As new observations, discoveries, and understandings emerge, classification methods should be revised.

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25. How did early classification methods differ from modern ones?
Early classification methods relied on external structures and the utility of organisms to categorize them. Modern classification, however, considers not just physical characteristics, but also evolutionary relationships and genetic characteristics to determine how organisms are classified.

26. According to modern classification methods, why do Salvinia and Wolffia belong to different families? How is Wolffia related with Anthurium?
Salvinia and Wolffia belong to different families because they have different evolutionary histories. Salvinia is a non-flowering plant in the family Salviniaceae, while Wolffia is a flowering plant in the family Araceae. Although they may seem similar in some ways, their genetic and evolutionary relationships are distinct.
Wolffia and Anthurium are included in the same family (Araceae) because they are believed to have evolved from a common ancestor.

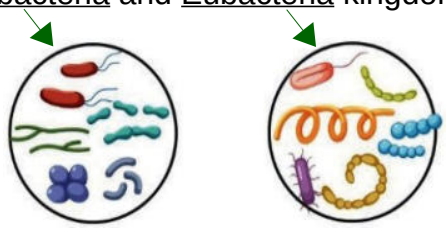
27. Identify the given plants. Which of these belong to the same family?



A- Azola
B- Wolfiya
C- Salvinia

Of these, azola and salvinia belong to the family, salviniacea.

28. Who proposed the six kingdom classification and how?
Carl Woese.
He divided Monera into Archebacteria and Eubacteria kingdoms. And the **6** kingdoms, as
- Archaebacteria,
- Eubacteria,
- Protista,
- Fungi,
- Plantae
- Animalia.



29. Why all humans belong to the same species, *sapiens*?
Although humans living in different parts of the earth show some differences in their external appearance due to the influence of local geography and environmental factors, all humans share a common ancestor. That is why all humans are classified under the same species, *sapiens*.

Video class links:
Part 1 :
Part 2 :

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